COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 010	SECTION: CPE21S4	
DATE PERFORMED:	26/07/2026	DATE SUBMITTED:	26/07/2026
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?

main.cpp

```cpp
#include <iostream>
using namespace std;

class Node {
public:
    char data;
    Node* next;
};

class LinkedList {
public:
    Node* head;

    LinkedList() {
        head = NULL;
    }

```

```
18      void insertAtEnd(char data) {
19          Node* newNode = new Node();
20          newNode->data = data;
21          newNode->next = NULL;
22
23          if (head == NULL) {
24              head = newNode;
25              return;
26          }
```

```
28          Node* temp = head;
29          while (temp->next != NULL) {
30              temp = temp->next;
31          }
32          temp->next = newNode;
33      }
34
35      void traverse() {
36          Node* temp = head;
37          while (temp != NULL) {
```

Output

```
C -> P -> E -> 0 -> 1 -> 0 -> NULL


=== Code Execution Successful ===
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: Copilot

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: I used it for helping me to fix my code to work on (Programiz)

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: It help me to fit the code on Programiz website because my VScode still not working.

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.

Supplementary

main.cpp

Share Run

```cpp
#include <iostream>
using namespace std;

class Node {
public:
    string songName;
    Node* next;
};

class Playlist {
public:
    Node* head;

    Playlist() { head = NULL; }

    void addSong(string name) {
        Node* newNode = new Node();
        newNode->songName = name;
        if (head == NULL) {
            head = newNode;
            newNode->next = head;
            return;
        }
        Node* temp = head;
        while (temp->next != head)
            temp = temp->next;
        temp->next = newNode;
        newNode->next = head;
    }

    void playAll() {
        if (head == NULL) {
```

main.cpp

Share Run

```
            return;
        }
        Node* temp = head;
        while (temp->next != head)
            temp = temp->next;
        temp->next = newNode;
        newNode->next = head;
    }

    void playAll() {
        if (head == NULL) {
            cout << "Playlist is empty!" << endl;
            return;
        }
        Node* temp = head;
        do {
            cout << temp->songName << " -> ";
            temp = temp->next;
        } while (temp != head);
        cout << "(back to start)" << endl;
    }
};

int main() {
    Playlist playlist;
    playlist.addSong("Song A");
    playlist.addSong("Song B");
    playlist.addSong("Song C");
    playlist.playAll();
    return 0;
}
```

Output

```
Song A -> Song B -> Song C -> (back to start)


=== Code Execution Successful ===
```

Conclusion: I learned how linked lists work in C++ and how nodes connect with pointers. At first it was confusing, but using Programiz helped me test the code step by step and see the output clearly. The procedure showed me how to create, insert, delete, and print nodes, and I realized that small pointer errors can break the program.

The playlist activity using a circular linked list was fun because it showed how coding can be used in real life, such as looping songs.

Overall, I think I did okay. I got the right output in Programiz, but I still get confused with pointers sometimes. I need more practice to improve, but I learned a lot from this activity.